

Element I: Evaluating Performance of Second Prototype

Test Results

- 1) Passed. (3/5) Able to mostly easily turn on and off while worn.
- 2) Passed. (4/5) Able to put on and take off belt
- 3) Passed. (3/5) durable, able to still work after being dropped.

Test

These 3 basic tests show that this prototype will work on a belt. This prototype just made of duct tape can work basic functions so we know that a more professional final product should also succeed in these basic functions. After analyzing this data we feel comfortable with continuing with our model but perhaps just make it a tiny small one so it is less obvious to see while worn.

Test 1) Activated Tracking

(Medium priority)

Testing Process ~ Take belt with tracker in tracking pocket. Put on and wear like a normal belt. Hold the button for 5 seconds to turn on and off.

Summary of Data ~ It was able to relatively easily turn on and off.

Confirmation or changes to project ~ This confirmed the process of using a pocket and being able to operate and remain comfortable.

Test 2) Belt Fitting

(Medium priority)

Testing Process ~ Take belt with tracker in tracking pocket. Put on and wear like a normal belt.

Summary of Data ~ Felt like a normal belt without any discomfort and also not covering the tracker but making it less obvious.

Confirmation or changes to project ~ This confirmed that it will work on a belt. No need to change or put comfort padding on the belt.

Test 3) Tracker Endurance

(Low priority)

Testing Process ~ Take belt with tracker in tracking pocket. Hold up to waist height and drop onto a hard floor.

Summary of Data ~ We dropped the belt 3 times but everytime the belt and the tracker were not damaged at all.

Confirmation or changes to project ~ It once again showed that the belt with the tracker can survive from small falls or light damage.